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## Integrated clustering system


### Abstract

Disclosed is an integrated clustering system including a master node that sets environment setting for clustering and generate an access token by installing a management program, a plurality of worker nodes that set environment setting based on an environment variable including an IP address of the master node and the access token and perform clustering with the master node by installing the management program, and a system control unit that extracts runtimes of clustering groups including the master node and the plurality of worker nodes which are clustered, and compares the runtime of one of the clustering groups with the runtimes of other clustering group than the one clustering group.



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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS
(1) This application claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2021-0127379 filed on Sep. 27, 2021, in the Korean Intellectual Property Office, the disclosures of which are incorporated by reference herein in their entireties.
BACKGROUND
(2) Embodiments of the present disclosure described herein relate to an integrated clustering system for automatically clustering a plurality of IoT devices.
(3) In general, many applications are being developed to process massive data generated from

sensors, actuators, and devices corresponding to IoT devices. In this case, Kubernets (K8s) is used to effectively manage and orchestrate numerous applications used in IoT devices.

(4) The Kubernets (K8s) requires system requirements that are too high to be installed and operated in IoT devices with limited resources, so technologies such as the Rancher K3s program are being utilized these days. However, to cluster a plurality of IoT devices using the technologies as the Rancher K3s program, there is inconvenience in that the user has to manually configure and install the requirements of the K3s program.

(5) The Kubernets (K8s) requires too high system requirements to be installed and operated in IoT devices with limited resources, so technologies such as the Rancher K3s program are being utilized these days. However, to cluster a plurality of IoT devices using the technologies as the Rancher K3s program, there is inconvenience in that the user has to manually configure and install the requirements of the K3s program.

(6) Accordingly, there is a need for a technology for installing the K3s program in IoT devices and automatically performing clustering between IoT devices.

SUMMARY

(7) Embodiments of the present disclosure provide an integrated clustering system for automatically clustering a plurality of IoT devices.

(8) According to an embodiment, an integrated clustering system includes at least one processor, wherein the at least one processor includes a master node that sets first environment setting based on a first environment variable for clustering and generate an access token by installing a management program, a plurality of worker nodes that set second environment setting based on a second environment variable including an IP address of the master node and the access token and perform clustering with the master node by installing the management program, and a system control unit that extracts runtimes of clustering groups including the master node and the plurality of worker nodes which are clustered, and compares the runtime of one of the clustering groups with the runtimes of other clustering group than the one clustering group.

(9) Further, the system control unit may set a first runtime during which a container provided in any one worker node among the plurality of worker nodes operates and cluster the master node and each of the plurality of worker nodes by setting the first runtime to the runtime during which a container provided in each of the other worker nodes than the any one worker node operates.

(10) Further, the system control unit may reinstall the management program by resetting the first environment variable, and perform the clustering of the master node and each of the plurality of worker nodes using the access token, when the runtime of at least one of the clustering groups is not identical.

(11) Further, the system control unit may determine that the master node and each of the plurality of worker nodes are clustered when the runtimes of the clustering groups are identical.

(12) Further, the master node may include a scheduler unit that generates pod data including the IP address of the master node, the access token, and port information of the plurality of worker nodes based on the first environment variable of the master node and a first tunnel proxy unit that wirelessly transmits the container including the pod data.

(13) Further, the master node may include a scheduler unit that generates pod data including the IP address of the master node, the access token, and port information of the plurality of worker nodes based on the first environment variable of the master node and a first tunnel proxy unit that wirelessly transmits the container including the pod data.

(14) Further, the system control unit may activate swap memories of the master node and the plurality of worker nodes.

(15) According to an embodiment, a method of controlling an integrated clustering system including a master mode, a plurality of worker nodes and a system control unit which are performed by at least one processor, the method includes setting, by the mater node, first environment setting based on a first environment variable for performing clustering and generating

an access token by using a management program, performing, by each of the plurality of worker nodes, second environment setting based on a second environment variable including an IP address of the master node and the access token and performing clustering with the master node by installing the management program, and extracting, the system control unit, runtimes of clustering groups each including the master node and the plurality of worker nodes which are clustered, and comparing the runtime of one of the clustering groups with the runtimes of other clustering group than the one clustering group.
(16) Further, the comparing of the runtime of one of the clustering groups with the runtimes of other clustering group than the one clustering group may include setting, by the system control unit, a first runtime during which a container provided in any one worker node among the plurality of worker nodes operates; and clustering, by the system control unit, the master node and each of the plurality of worker nodes by setting the first runtime to the runtime during which a container provided in each of the other worker nodes than the any one worker node operates.
(17) Further, the extracting of the runtimes of clustering groups each including the master node and each of the plurality of worker nodes may include reinstalling the management program by resetting the first environment variable, and performing the clustering of the master node and each of the plurality of worker nodes using the access token, when the runtime of at least one of the clustering groups is not identical.
(18) According to an embodiment, a computer-readable non-transitory recording medium recording a program for executing the method of controlling the integrated clustering system according to the present disclosure is provoded.

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## Description

BRIEF DESCRIPTION OF THE FIGURES
(1) The above and other objects and features of the present disclosure will become apparent by describing in detail embodiments thereof with reference to the accompanying drawings.
(2) FIG. **1** is a diagram for describing an integrated clustering system according to an embodiment of the present disclosure;
(3) FIG. **2** is a diagram for describing the components of a master node and a worker node according to an embodiment of the present disclosure;
(4) FIG. **3** is a diagram for describing a process of setting the environment of a master node and a worker node according to an embodiment of the present disclosure;
(5) FIG. **4** is a diagram for describing a process of installing a management program on a master node according to an embodiment of the present disclosure;
(6) FIG. **5** is a diagram for controlling a process of installing a management program on a worker node according to an embodiment of the present disclosure; and
(7) FIG. **6** is a diagram for describing a clustering process according to an embodiment of the present disclosure.
DETAILED DESCRIPTION
(8) Hereinafter, with reference to the accompanying drawings, various embodiments of the present disclosure will be described in detail such that those of ordinary skill in the art can easily carry out the present disclosure. The present disclosure may be embodied in several different forms and is not limited to the embodiments described herein.
(9) In order to clearly explain the present disclosure, parts irrelevant to the present disclosure are omitted, and the same reference numerals are assigned to the same or similar elements throughout the specification. Accordingly, the reference numerals described above may be used in other drawings as well.
(10) Further, in the drawings, a size, and a thickness of each element are arbitrarily illustrated for

convenience of description, and the present disclosure is not necessarily limited to those illustrated in the drawings. In the drawings, the thicknesses of layers and regions are enlarged for clarity.
(11) The expression “the same” means to “substantially the same”. That is, it may be the same degree to the extent that ordinary knowledge those of ordinary skill in the art can convince that they are the same. Other expressions may be expressions in which “substantially” is omitted.
(12) FIG. **1** is a diagram for describing an integrated clustering system according to an embodiment of the present disclosure.
(13) The integrated clustering system (K3s clustering system) **1** according to an embodiment of the present disclosure may include at least one processor.
(14) The at least one processor may include a master node **10**, a worker node **20**, a system control unit **30**, and a management program installation unit (K3s program unit) **40**.
(15) The master node **10** may be clustered with a plurality of worker nodes **20**, individually. Clustering refers to connection or coupling of any one master node **10** for controlling a plurality of IoT devices (or worker nodes **20**) and a plurality of worker nodes **20**. That is, the master node **10** may be clustered with each of the plurality of worker nodes **20** to orchestrate or control the plurality of worker nodes **20**.
(16) The master node **10** may be subjected to environment setting based on variables for clustering. In this case, the environment setting refers to setting the environment of the master node **10** necessary for the master node **10** to cluster with each of the plurality of worker nodes **20**. Hereinafter, the variables for setting an environment in the master node **10** are referred to as first environment variables, and it is assumed that the master node **10** is subjected to first environment setting based on the first environment variables.
(17) A management program for performing clustering with each of the plurality of worker nodes **20** may be installed in the master node **10** on which the first environment setting has been completed. The master node **10** may be clustered with each of the plurality of worker nodes **20** using a management program.
(18) Specifically, the master node **10** may have a management program installed, and generate an access token, which is a medium for performing clustering with the plurality of worker nodes **20**, based on the management program. The master node **10** may be clustered with each of the plurality of worker nodes **20** by using the access token generated by the master node **10** as a medium for performing clustering with each of the plurality of worker nodes **20**.
(19) Each of the plurality of worker nodes **20** may be subjected to environment setting using variables including the IP address of the master node **10**, the access token generated in the master node **10**, and port information of each of the plurality of worker nodes **20**. That is, since the environment of the plurality of worker nodes **20** is set based on the IP address of the master node **10**, the access token, and the port information of the plurality of worker nodes **20**, each of the plurality of worker nodes **20** may be clustered with the master node **10**. Hereinafter, a variable for performing environment setting in each of the plurality of worker nodes **20** is referred to as a second environment variable, and it is assumed that the plurality of worker nodes **20** are subjected to second environment setting based on the second environment variable.
(20) The system control unit **30** may calculate a first environment variable and a second environment variable for clustering the master node **10** and the plurality of worker nodes **20**, respectively. The system control unit **30** may perform control such that the first environment setting and the second environment setting of the master node **10** and the plurality of worker nodes **20** are completed based on the first environment variable and the second environment variable.
(21) The system control unit **30** may perform control such that the master node **10** in which the management program is installed and each of the plurality of worker nodes **20** are clustered. Specifically, the system control unit **30** may set a runtime during which a container **28** (see FIG. **2**) provided in one among the plurality of worker nodes **20** operates to a first runtime, which is a reference runtime. The system control unit **30** may set the first runtime during which the containers

**28**, respectively provided in the plurality of worker nodes other than the one worker node, operate as a first runtime.

(22) The system control unit **30** may detect the operation of the container **28** (refer to FIG. **2**) operating in the plurality of clustered worker nodes **20** to determine whether clustering of the master node **10** and each of the plurality of worker nodes **20** is performed.

(23) Specifically, the system control unit **30** may measure the runtimes of the containers **28** operating in the plurality of clustered worker nodes **20**. When the runtime of at least one of the plurality of clustered worker nodes **20** is not identical, the system control unit **30** may determine that the master node **10** and each of the plurality of worker nodes **20** have not been clustered.

(24) In this case, the system control unit **30** may again set the first environment variable of the master node **10** to reinstall the management program. The system control unit **30** may re-perform the clustering of the master node **10** and each of the plurality of worker nodes **20** by using a newly-generated access token.

(25) When all the runtimes of the clustering group are the same, the system control unit **30** may determine that the master node **10** and each of the plurality of worker nodes **20** are clustered.

(26) In this case, the system control unit **30** may perform control such that the clustering of the master node **10** and the plurality of worker nodes **20** are maintained.

(27) The system control unit **30** may deactivate a swap memory **25** (see FIG. **2**) included in each of the plurality of worker nodes **20**. The system control unit **30** may perform control such the data is stored only in a memory unit **26** of the plurality of worker nodes **20** by inactivating the swap memory **25**.

(28) A process in which the system control unit **30** determines the first environment variable and the second environment variable to cluster the master node **10** and each of the plurality of worker nodes **20**, and performs control to complete the first environment setting and the second environment setting of the master node **10** and the plurality of worker nodes **20** based on the first environment variable and the second environment variable and a process in which in which the system control unit **30** performs control such that the master node **10** and each of the plurality of worker nodes **20** are clustered will be described in detail below with reference to FIGS. **2** and **3**.

(29) In addition, a process in which the system control unit **30** determines whether to perform clustering of the master node **10** and each of the plurality of worker nodes **20** using data about a plurality of clustering groups in which the management program is installed will be described in detail below with reference to FIG. **6**.

(30) The management program installation unit **40** may store a management program to be installed in the master node **10** on which the first environment setting has been completed. The management program installation unit **40** may provide a management program to each of the plurality of worker nodes **20** on which the second environment setting has been completed. In this case, the management program refers to a program necessary to perform clustering of one master node **10** and the plurality of worker nodes **20**, and may include Rancher K3s, or the like, but the present disclosure is not limited thereto.

(31) FIG. **2** is a diagram for describing the components of a master node and a worker node according to an embodiment of the present disclosure.

(32) The master node **10** may include an API server unit **11**, a database unit **12**, a scheduler unit **13**, and a first tunnel proxy unit **14**.

(33) The worker node **20** may include a second tunnel proxy unit **21**, a kube proxy unit **22**, a flannel unit **23**, the swap memory **25**, a kubelet unit **24**, the memory unit **26**, and a pod **27**.

(34) The second tunnel proxy unit **21**, the kube proxy unit **22**, the flannel unit **23**, the swap memory **25**, the kubelet unit **24**, the memory unit **26**, and the pod **27** may be performed by at least one processor.

(35) The API server unit **11** may perform first environment setting based on the first environment variable calculated by the system control unit **30**. The API server unit **11** may install a management

program provided by the management program installation unit **40**. The API server unit **11** may generate an access token, which is a medium for performing clustering, based on the management program. The API server unit **11** may perform clustering with each of the plurality of worker nodes **20** by using the access token.

(36) The database unit **12** may store the first environment variable. The database unit **12** may store data related to the management program. The database unit **12** may store the access token, which is a medium for performing clustering.

(37) The scheduler unit **13** may generate pod data including the IP address of the master node **10**, the access token, and the port information of the plurality of worker nodes **20**. In this case, the pod data generated by the scheduler unit **13** may be stored in each of the containers **28** provided in the pod **27** of the worker node **20** through wireless data transmission. The scheduler unit **13** may perform scheduling such that the pod **27** is allocated to at least one worker node **20** according to a user's selection.

(38) The first tunnel proxy unit **14** may perform wireless data transmission between the master node **10** and the worker node **20**. The first tunnel proxy unit **14** may perform wireless data transmission to the second tunnel proxy unit **21** of the worker node **20**. The first tunnel proxy unit **14** may wirelessly transmit, to the second tunnel proxy unit **21** of the worker node **20**, the pod data including the IP address of the master node **10**, the access token, and the port information of the plurality of worker nodes **20**.

(39) The second tunnel proxy unit **21** may perform wireless data reception between the worker node **20** and the master node **10**. The second tunnel proxy unit **21** may wirelessly receive data from the first tunnel proxy unit **14** of the master node **10**. The second tunnel proxy unit **21** may wirelessly receive, from the first tunnel proxy unit **14** of the master **10**, the pod data including the IP address of the master node **10**, the access token, and the port information of the plurality of worker nodes **20**.

(40) The kube proxy unit **22** may perform a load balancing function. Load balancing may mean distributing the pod data transmitted wirelessly from the master node **10** to the plurality of worker nodes **20**. That is, the kube proxy unit **22** may perform load balancing such that pod data corresponding to the capacity of the memory unit **26** is distributed to the plurality of worker nodes **20**.

(41) The flannel unit **23** may support network communication between the pods **27** provided in the plurality of worker nodes **20**. The flannel unit **23** may support network communication between the containers **28** provided in the plurality of worker nodes **20**. The flannel unit **23** may support network communication between the pods **27** and the containers **28** provided in the plurality of worker nodes **20**.

(42) The kubelet unit **24** may drive the pod **27** in the worker node **20**. The kubelet unit **24** may drive the containers **28** included in the worker node **20**. The kubelet unit **24** may install the management program in the worker node **20** by driving the pod data included in the containers **28** and management program data.

(43) The pod **27** may include the plurality of containers **28**. The containers **28** included in the pod **27** may include data related to the management program. The containers **28** included in the pod **27** may include pod data. In this case, in the containers **28** included in the pod **27**, environment may be set based on the pod data, and the management program may be executed and installed.

(44) That is, the plurality of worker nodes **20** may set the second environment variable based on the pod data provided in the containers **28** and perform the second environment setting. A management program may be installed in the plurality of worker nodes **20** on which the second environment setting has been completed.

(45) The memory unit **26** may store data related to the management program provided in the containers **28** and the pod data. The swap memory **25** is deactivated by the system control unit **30**, so that pod data and/or management programs are not stored.

(46) FIG. **3** is a diagram for describing a process of setting the environment of a master node and a worker node according to an embodiment of the present disclosure.
(47) The master node **10**, the plurality of worker nodes **20**, and the system control unit **30** according to an embodiment of the present disclosure may be executed by at least one processor.
(48) In step S**10**, the system control unit **30** may activate an IP table of the master node **10**.
(49) Specifically, since each of the plurality of worker nodes **20** performs the second environment setting using the second environment variable including the IP address of the master node **10** as described above with reference to FIG. **1**, it is needed to access the IP table of the master node **10**. Accordingly, the system control unit **30** may activate the IP table of the master node **10** to access the IP address of the master node **10**.
(50) In step S**11**, the system control unit **30** may perform control such that access to the port list of each of the plurality of worker nodes **20** is possible.
(51) Specifically, since each of the plurality of worker nodes **20** performs the second environment setting using the second environment variable including the port information of each of the plurality of worker nodes **20**, it may be needed to access the port information of each of the plurality of worker nodes **20**. Accordingly, the system control unit **30** may perform control such that access to the port list of each of the plurality of worker nodes **20** is possible so as to calculate port information of each of the plurality of worker nodes **20**.
(52) In step S**12**, the system control unit **30** may deactivate the swap memory **25** of each of the plurality of worker nodes **20**.
(53) Specifically, the system control unit **30** may deactivate the swap memory **25** included in each of the plurality of worker nodes **20** such that data is stored only in the memory unit **26** included in each of the plurality of worker nodes **20**.
(54) In step S**13**, the system control unit **30** may activate Cgroups (control groups) of the master node **10** and the plurality of worker nodes **20**.
(55) Specifically, the system control unit **30** may activate the kube proxy unit **22** included in each of the plurality of worker nodes **20** to perform a load balancing function. The kube proxy unit **22** may perform load balancing such that pod data corresponding to the capacity of the memory unit **26** is distributed to the plurality of worker nodes **20**.
(56) In step S**14**, the system control unit **30** may set a runtime, which is a time during which the containers **28** provided in any one worker node among the plurality of worker nodes **20** operate.
(57) Specifically, the system control unit **30** may determine any one of the plurality of worker nodes **20** as a reference worker node and set the first runtime during which any one of containers **28** provided in the reference worker node operates, as a reference runtime.
(58) In addition, the system control unit **30** may set the first runtime, which is the time during which any one of the containers **28** provided in the reference worker node operates, as a runtime during which the containers **28** provided in the plurality of worker nodes other than the reference worker node operate.
(59) FIG. **4** is a diagram for describing a process of installing a management program on a master node according to an embodiment of the present disclosure.
(60) In step S**20**, the system control unit **30** may set an environment variable for clustering the master node **10** and the worker node **20**.
(61) Specifically, the system control unit **30** may determine a first environment variable for clustering the master node **10** and each of the plurality of worker nodes **20**.
(62) In this case, the master node **10** may perform first environment setting based on the first environment variable.
(63) In step S**21**, the management program installation unit **40** may install the management program on the master node **10**.
(64) Specifically, the management program installation unit **40** may store the management program to be installed in the master node **10** on which the first environment setting has been completed.

The management program installation unit **40** may provide the management program to the master node **10** on which the first environment setting has been completed. The master node **10** on which the first environment setting has been completed may install the management program provided from the management program installation unit **40**.
(65) In step S**22**, the master node **20** may generate an access token.
(66) The master node **10** on which the management program has been installed may generate an access token, which is a medium for performing clustering with the plurality of worker nodes **20**.
(67) In step S**23**, the master node **10** may send pod data including the IP address of the master node **10**, the access token, and port information of each of the plurality of worker nodes **20** to each of the plurality of worker nodes **20**.
(68) Specifically, the API server unit **11** provided in the master node **10** may generate an access token, which is a medium for performing clustering, based on the management program installed in the master node **10**.
(69) The scheduler unit **13** provided in the master node **10** may generate pod data including the IP address of the master node **10**, the access token, and the port information of the plurality of worker nodes **20**. In this case, the pod data generated by the scheduler unit **13** may be stored in each of the containers **28** provided in the pod **27** of the worker node **20** through wireless data transmission.
(70) FIG. **5** is a diagram for controlling a process of installing a management program on a worker node according to an embodiment of the present disclosure.
(71) In step S**30**, a second environment variable is set in the plurality of worker nodes **20** based on the pod data including the IP address of the master node **10**, the access token, and the port information and the plurality of worker nodes **20** each may be subjected to second environment setting.
(72) In step S**31**, the management program installation unit **40** may install a management program on the plurality of worker nodes **20**.
(73) Specifically, the management program installation unit **40** may store the management program to be installed in each of the plurality of worker nodes **20** on which the second environment setting has been completed. The management program installation unit **40** may provide a management program to the plurality of worker nodes **20** on which the second environment setting has been completed. Each of the plurality of worker nodes **20** in which the second environment setting has been completed may install a management program provided from the management program installation unit **40**.
(74) In step S**32**, each of the plurality of worker nodes **20** may perform clustering with the master node **10**.
(75) The plurality of worker nodes **20** in which the second environment setting is completed and the management program is installed may perform clustering with the master node **10**.
(76) FIG. **6** is a diagram for describing a clustering process according to an embodiment of the present disclosure.
(77) In step S**40**, the system control unit **30** may calculate the runtime of each clustering group.
(78) Specifically, the system control unit **30** may calculate the runtimes of the containers **28** included in each of the plurality of worker nodes **20** of a clustering group that is clustered to operate.
(79) In step S**41**, the system control unit **30** may determine whether the runtimes of the plurality of worker nodes **20** of the clustering group are identical to each other.
(80) When the runtimes of the plurality of worker nodes **20** of the clustering group are all the same in step S**42**, the system control unit **30** may determine that the master node **10** and each of the plurality of worker nodes **20** has been clustered.
(81) When the runtime of at least one of the worker nodes **20** of the clustering group is not identical, in step S**43**, the system control unit **30** may determine that the master node **10** and each of the plurality of worker nodes **20** have not been clustered.

(82) the system control unit **30** may again set the first environment variable of the master node **10** to reinstall the management program. In addition, the system control unit **30** may re-perform the clustering of the master node **10** and each of the plurality of worker nodes **20** by using an access token newly generated in the master node **10**.
(83) The integrated clustering system according to the present disclosure is able to automatically cluster a plurality of IoT devices.
(84) The drawings and the detailed description of the present disclosure as described above are merely examples of the present disclosure, which are only used for the purpose of explaining the present disclosure, and are not used to limit the meaning or the scope of the present disclosure described in the claims. Therefore, those skilled in the art will understand that various modifications and equivalent other embodiments are possible therefrom. Accordingly, the technical scope of the present disclosure should be defined by the accompanying claims.
(85) While the present disclosure has been described with reference to embodiments thereof, it will be apparent to those of ordinary skill in the art that various changes and modifications may be made thereto without departing from the spirit and scope of the present disclosure as set forth in the following claims.

## Claims

1. An integrated clustering system comprising at least one processor, wherein the at least one processor includes a master node configured to set first environment setting based on a first environment variable for clustering and generate an access token by installing a management program; a plurality of worker nodes configured to set second environment setting based on a second environment variable including an IP address of the master node and the access token and perform clustering with the master node by installing the management program; and a system control unit configured to extract runtimes of the plurality of worker nodes which are clustered with the master node in a clustering group, and compare the extracted runtimes of the plurality of worker nodes to each other, wherein the extracted runtimes refer to an execution environment.
2. The integrated clustering system of claim 1, wherein the system control unit is further configured to: set a first runtime during which a container provided in one of the plurality of worker nodes operates as a reference runtime; and cluster the master node and each of the plurality of worker nodes by setting the reference runtime as an operating runtime during which a container provided in each of worker nodes other than the one worker node operates.
3. The integrated clustering system of claim 2, wherein the system control unit is further configured to reinstall the management program by resetting the first environment variable, and perform the clustering of the master node and each of the plurality of worker nodes using the access token, when the extracted runtimes of the plurality of worker nodes are not identical.
4. The integrated clustering system of claim 2, wherein the system control unit is configured to determine that the master node and each of the plurality of worker nodes are clustered when the extracted runtimes of the plurality of worker nodes are identical.
5. The integrated clustering system of claim 2, wherein the master node includes: a scheduler unit configured to generate pod data including the IP address of the master node, the access token, and port information of the plurality of worker nodes based on the first environment variable of the master node; and a first tunnel proxy unit configured to wirelessly transmit the container including the pod data.
6. The integrated clustering system of claim 5, wherein each of the plurality of worker nodes includes a second tunnel proxy unit configured to wirelessly transmit the container including the pod data, and wherein each of the plurality of worker nodes performs second environment setting using the IP address of the master node and the access token as the second environment variable and installs the management program based on the second environment setting.

7. The integrated clustering system of claim 6, wherein the system control unit is configured to deactivate swap memories of the master node and the plurality of worker nodes.
8. A method of controlling an integrated clustering system including a master node, a plurality of worker nodes and a system control unit which are performed by at least one processor, the method comprising: setting, by the master node, first environment setting based on a first environment variable for performing clustering and generating an access token by using a management program; performing, by each of the plurality of worker nodes, second environment setting based on a second environment variable including an IP address of the master node and the access token and performing clustering with the master node by installing the management program; and extracting, by the system control unit, runtimes of the plurality of worker nodes which are clustered with the master node in a clustering group, and comparing the extracted runtimes of the plurality of worker nodes to each other, wherein the extracted runtimes refer to an execution environment.
9. The method of claim 8, further comprising: setting, by the system control unit, a first runtime during which a container provided in one of the plurality of worker nodes operates as a reference runtime; and clustering, by the system control unit, the master node and each of the plurality of worker nodes by setting the reference runtime as an operating runtime during which a container provided in each of worker nodes other than the one worker node operates.
10. The method of claim 9, further comprising: reinstalling the management program by resetting the first environment variable, and performing the clustering of the master node and each of the plurality of worker nodes using the access token, when the extracted runtimes of the plurality of worker nodes are not identical.
11. A computer-readable non-transitory recording medium recording a program for executing the method of controlling the integrated clustering system according to claim 8.